

**CS 35201 Computer Communication Networks**  
**Spring 2025**  
**Department of Computer Science**  
**Kent State University**

**Final project**

---

**Sockets programming**

**Due date: April 28th**

1. Using the code provided with the homework, and the pygame python library, continue working on the client-server application of the “bucket catch” game.
2. The objective of the game is to collect as many falling objects as possible, similar to the “bucket catch” game.



3. Rules of the game:
  - a. One bucket moves sideways (even up and down) to collect all falling objects.
  - b. Falling objects must be generated at random locations from the top of the screen from the starting position on the X-axis.
  - c. Objects will fall faster as the game progresses.
  - d. Bucket also moves faster as the game progresses.
  - e. Player loses as soon as the first object touches the bottom of the screen.
4. A score must be displayed on the screen. No leaderboard is necessary.
5. GUI is not an important part of the homework, it is not necessary to implement any visual elements outside of the basic elements to execute the game properly.

6. The server side will be in charge of displaying the game while the client side will be in charge of the controls, you can set the controls to be with any keys from the keyboard, preferable with the "WASD" or arrow keys.
7. Homework can be developed in pairs of two students, allowing for 2 people to work on the 2 people's project made specifically for 2 people and no less or more than 2 people.
8. The project will be evaluated during class time by the instructor and the grader together. With students running the server and client side on two separate computers.
9. Project must be submitted to canvas as a .rar or .zip file containing all the source codes.

Evaluation table

| Tasks                                                                  | Points - 20 total |
|------------------------------------------------------------------------|-------------------|
| Client-server connection established                                   | 3 pts             |
| Client gets key input and sends it to server                           | 3 pts             |
| Server receives client data                                            | 3 pts             |
| Server side uses threads to manage game GUI and server                 | 2 pts             |
| Main bucket moves on the screen based on client's key press            | 2 pts             |
| Objects are randomly generated on the screen                           | 1 pt              |
| Objects fall at a faster speed as the game progresses                  | 1 pt              |
| Bucket moves faster as the game progresses                             | 1 pt              |
| Collision between "bucket" and falling objects                         | 1 pt              |
| Game ends when 1 <sup>st</sup> object reaches the bottom of the screen | 1 pt              |
| Functional score board                                                 | 1 pt              |
| Game can be restarted once it finishes                                 | 1 pt              |

Additional notes:

- a) The pygame and keyboard libraries must be installed alongside with python to develop the game.
- b) MacOS doesn't support the "keyboard" library, use pynput instead.
- c) You can use github to manage the development of the project.